3.

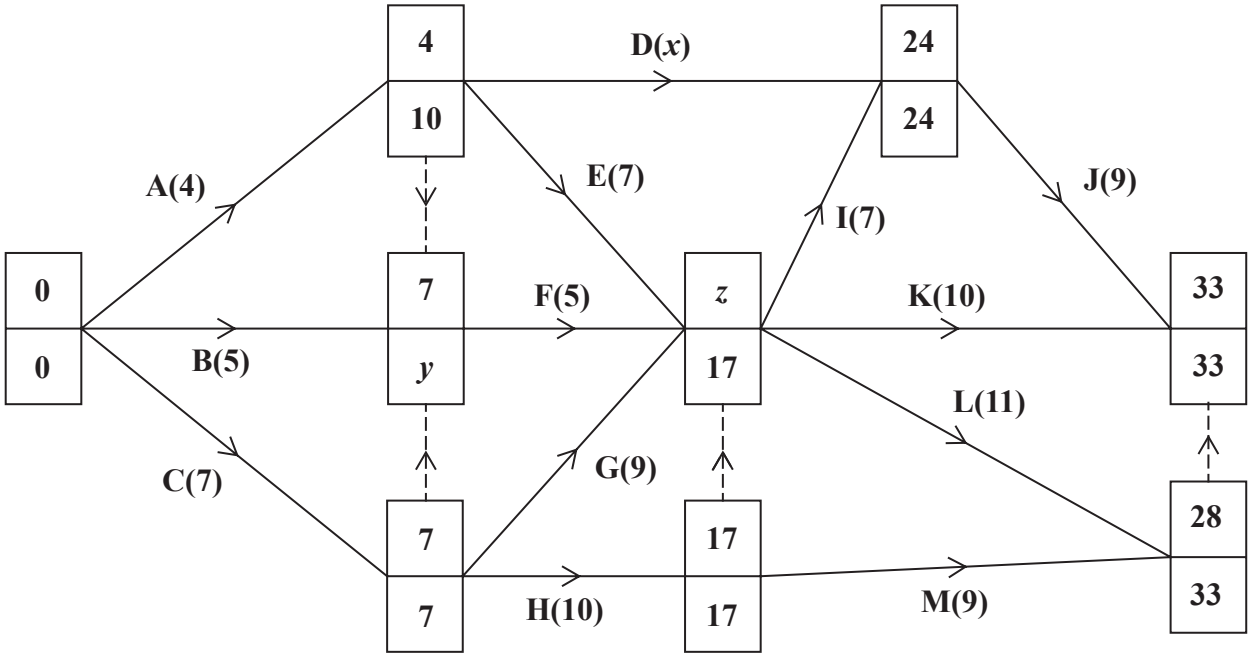


Figure 2

The network in Figure 2 shows the activities that need to be undertaken by a company to complete a project. Each activity is represented by an arc and the duration, in days, is shown in brackets. Each activity requires one worker. The early event times and late event times are shown at each vertex.

The total float on activity D is twice the total float on activity E.

(a) Find the values of x, y and z.

(3)

(b) Draw a cascade chart for this project on Grid 1 in the answer book.

(4)

(c) Use your cascade chart to determine a lower bound for the minimum number of workers needed to complete the project in the shortest possible time. You must make specific reference to time and activities. (You do not need to provide a schedule of the activities.)

(2)

(Total 9 marks)

Leave blank

3.

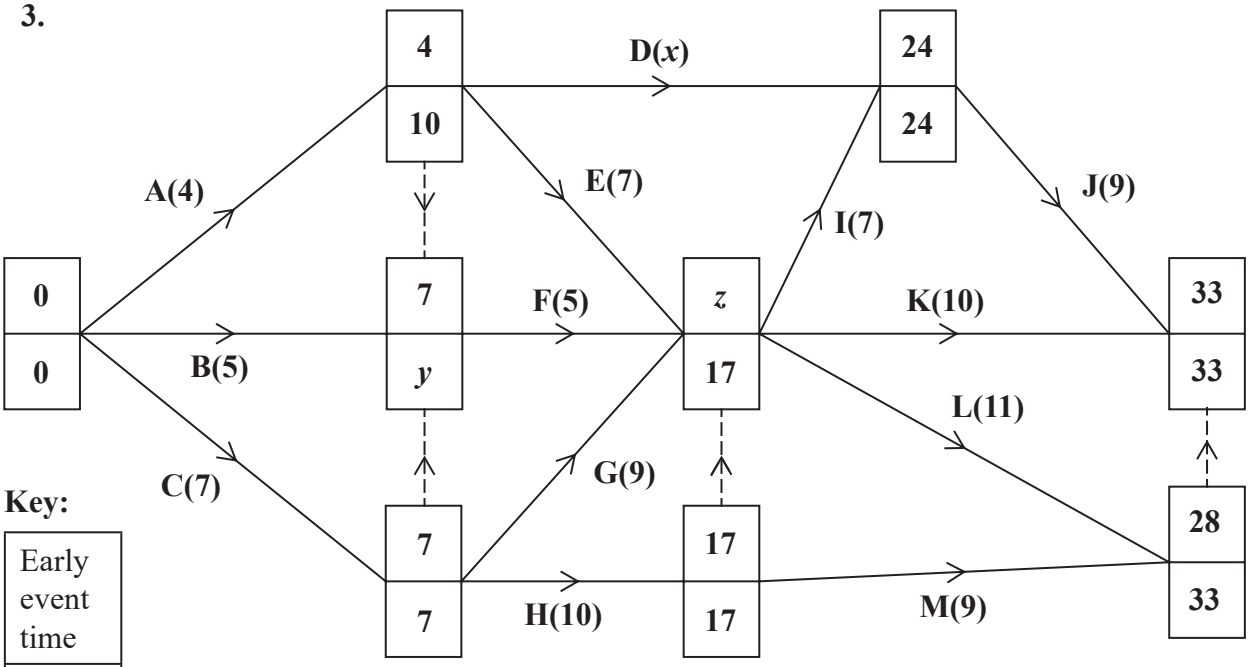


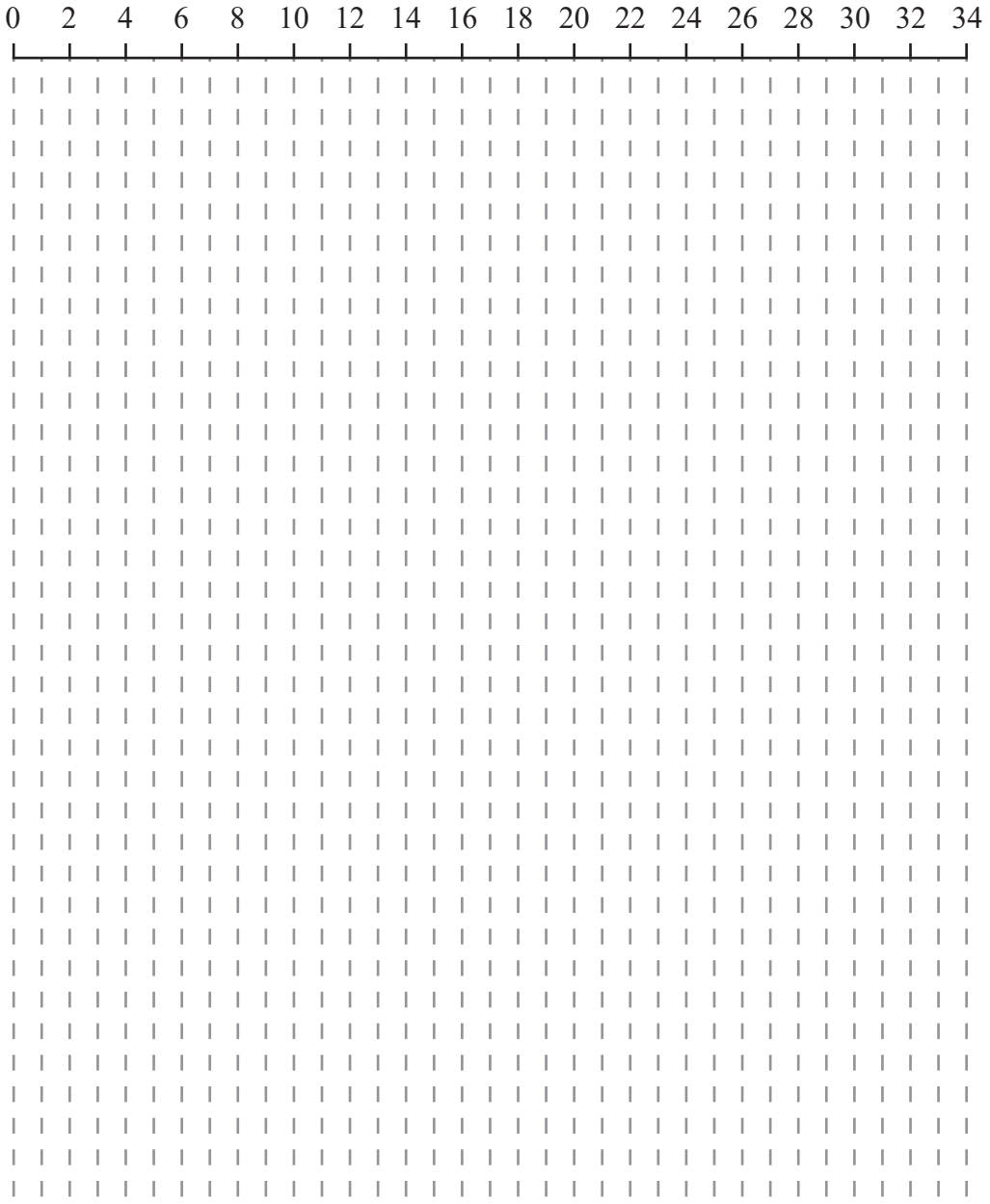
Figure 2

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Leave blank

Question 3 continued



Grid 1

Q3

(Total 9 marks)

5.

Activity	Immediately preceding activities
A	–
B	–
C	–
D	A
E	C
F	A, B, C
G	A, B, C
H	D, F, G
I	A, B, C
J	D, F, G
K	H
L	D, E, F, G, I

(a) Draw the activity network described in the precedence table above, using activity on arc. Your activity network must contain only the minimum number of dummies.

(5)

Given that all critical paths for the network include activity H,

(b) state which activities cannot be critical.

(2)

(Total 7 marks)

Leave blank

5.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Leave blank

Question 5 continued

(Total 7 marks)

Q5

6.

Activity	Duration (days)	Immediately preceding activities
A	4	–
B	7	–
C	6	–
D	10	A
E	5	A
F	7	C
G	6	B, C, E
H	6	B, C, E
I	7	B, C, E
J	9	D, H
K	8	B, C, E
L	4	F, G, K
M	6	F, G, K
N	7	F, G
P	5	M, N

The table above shows the activities required for the completion of a building project. For each activity the table shows the duration, in days, and the immediately preceding activities. Each activity requires one worker. The project is to be completed in the shortest possible time.

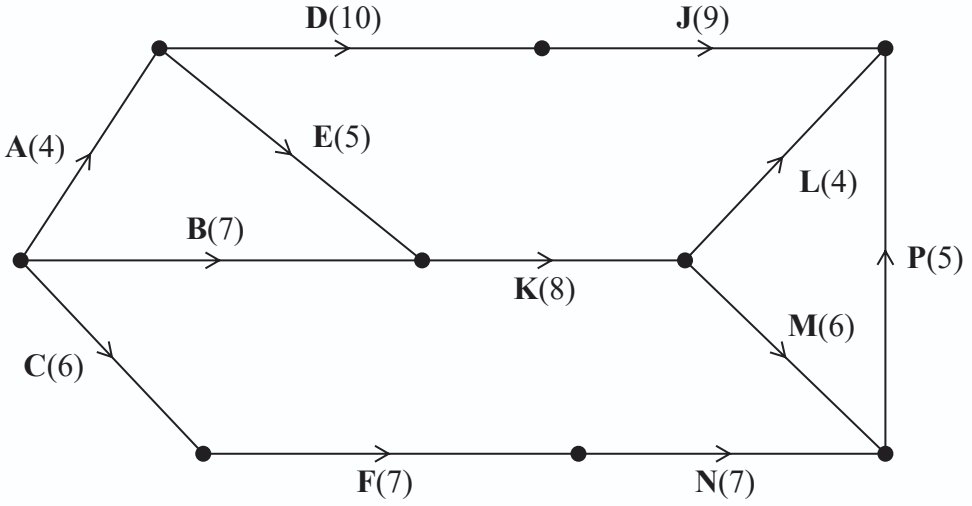


Figure 2

Figure 2 shows a **partially completed** activity network used to model the project. The activities are represented by the arcs and the numbers in brackets on the arcs are the times taken, in days, to complete each activity.

(a) Complete the network in Diagram 1 in the answer book by adding activities G, H and I and the minimum number of dummies.

(3)

(b) Add the early event times and the late event times to Diagram 1 in the answer book.

(4)

(c) State the critical activities.

(1)

(d) Calculate a lower bound for the number of workers needed to complete the project in the shortest possible time. You must show your working.

(2)

(e) Schedule the activities on Grid 1 in the answer book, using the minimum number of workers, so that the project is completed in the minimum time.

(3)

(Total 13 marks)

Leave blank

6. (a) and (b)

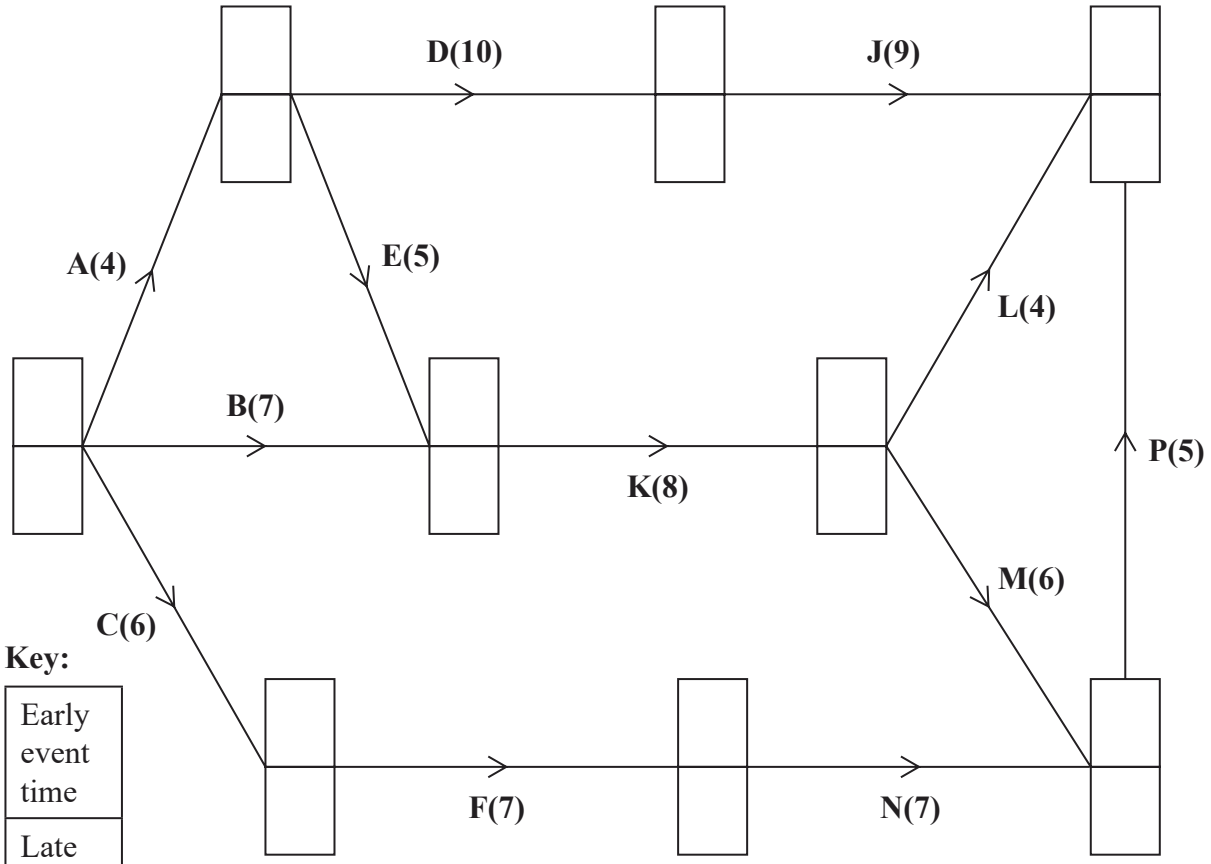


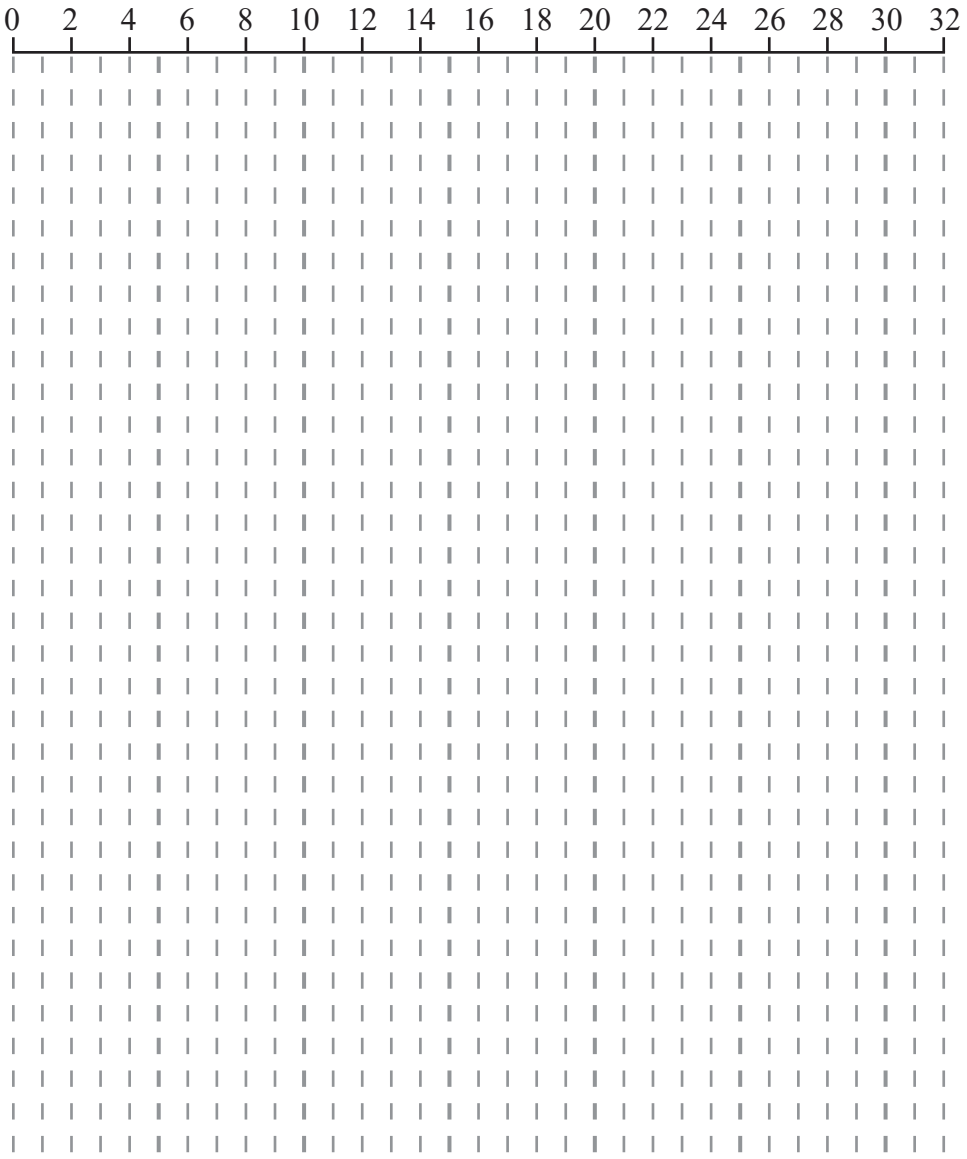
Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Leave blank

Question 6 continued



Grid 1

Q6

(Total 13 marks)

3.

Activity	Immediately preceding activities
A	–
B	–
C	–
D	A, B, C
E	A, B, C
F	C
G	F
H	D
I	D, E, G
J	D, E

Draw the activity network described in the precedence table above, using activity on arc and exactly 4 dummies.

(5)

(Total 5 marks)

3.

Activity	Immediately preceding activities
A	–
B	–
C	–
D	A, B, C
E	A, B, C
F	C
G	F
H	D
I	D, E, G
J	D, E

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 3 continued

Please redraw your activity network on this page if you need to do so.

(Total 5 marks)

4.

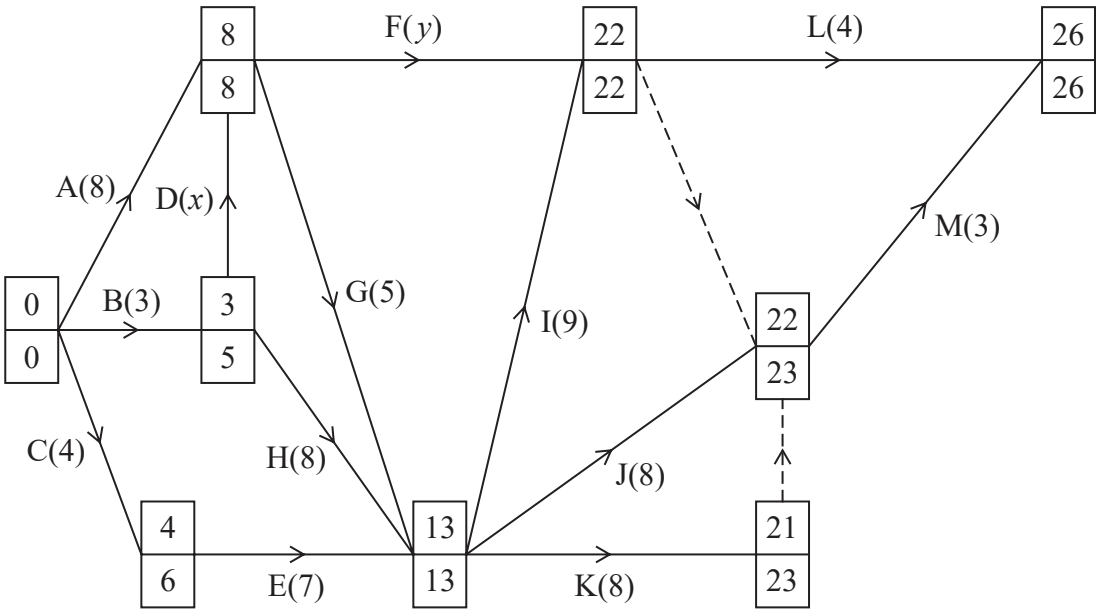


Figure 3

The network in Figure 3 shows the activities that need to be undertaken by a company to complete a project. Each activity is represented by an arc. The duration of the activity, in days, is shown in brackets. Each activity requires exactly one worker. The early event times and the late event times are shown at the vertices.

It is given that the total float on activity F is twice the total float on activity D.

It is also given that the total duration of the activities on the path BDFM is 10 days less than the duration of the critical path.

(a) Determine the value of x and the value of y. You must make your method and working clear.

(4)

(b) Draw a cascade chart for this project on Grid 1 in the answer book.

(4)

(c) Use your cascade chart to determine the minimum number of workers needed to complete the project in the shortest possible time. You must make specific reference to time and activities. (You do not need to provide a schedule of the activities.)

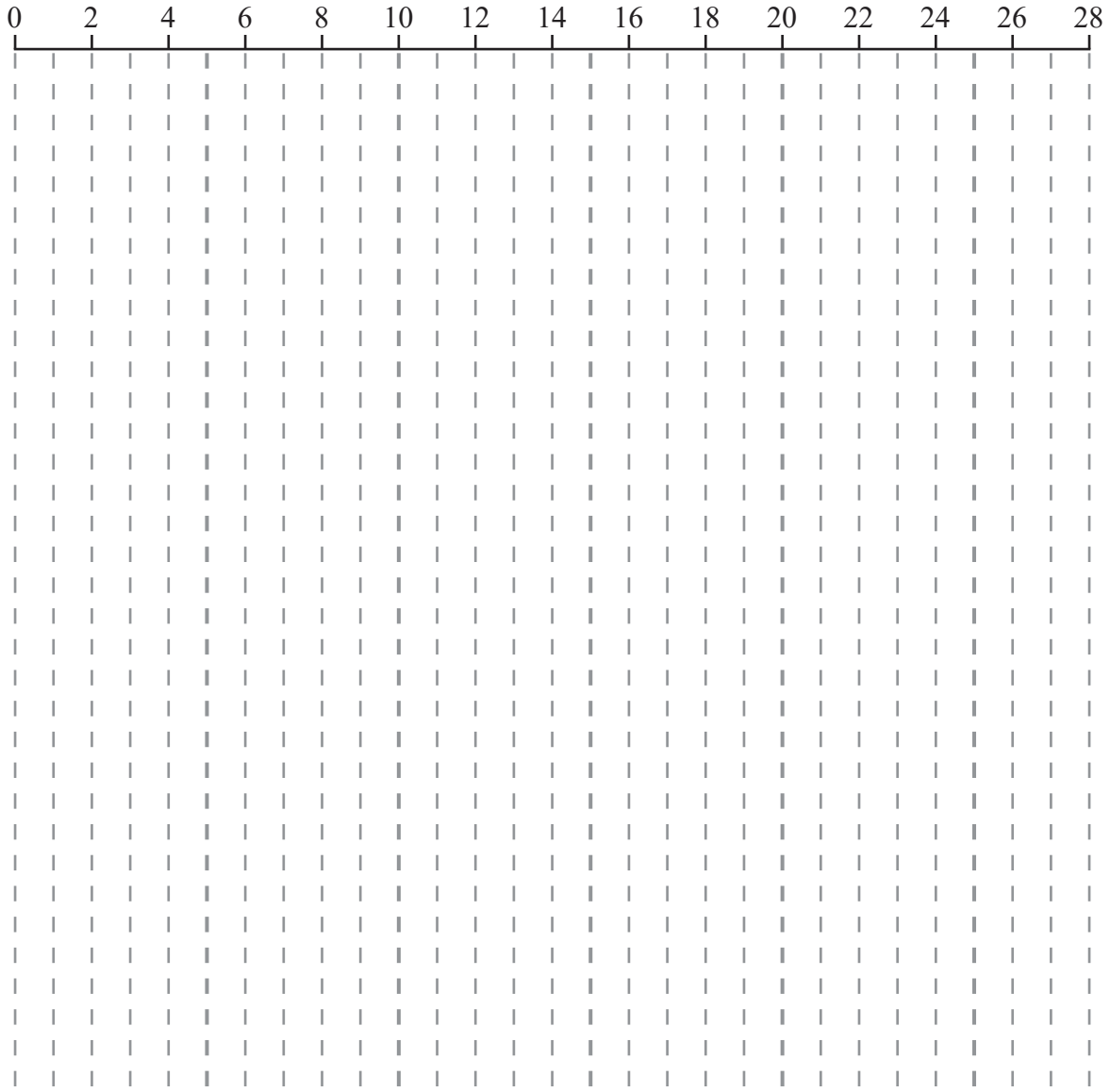
(2)

(Total 10 marks)

4.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 4 continued



Grid 1

(Total 10 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4.

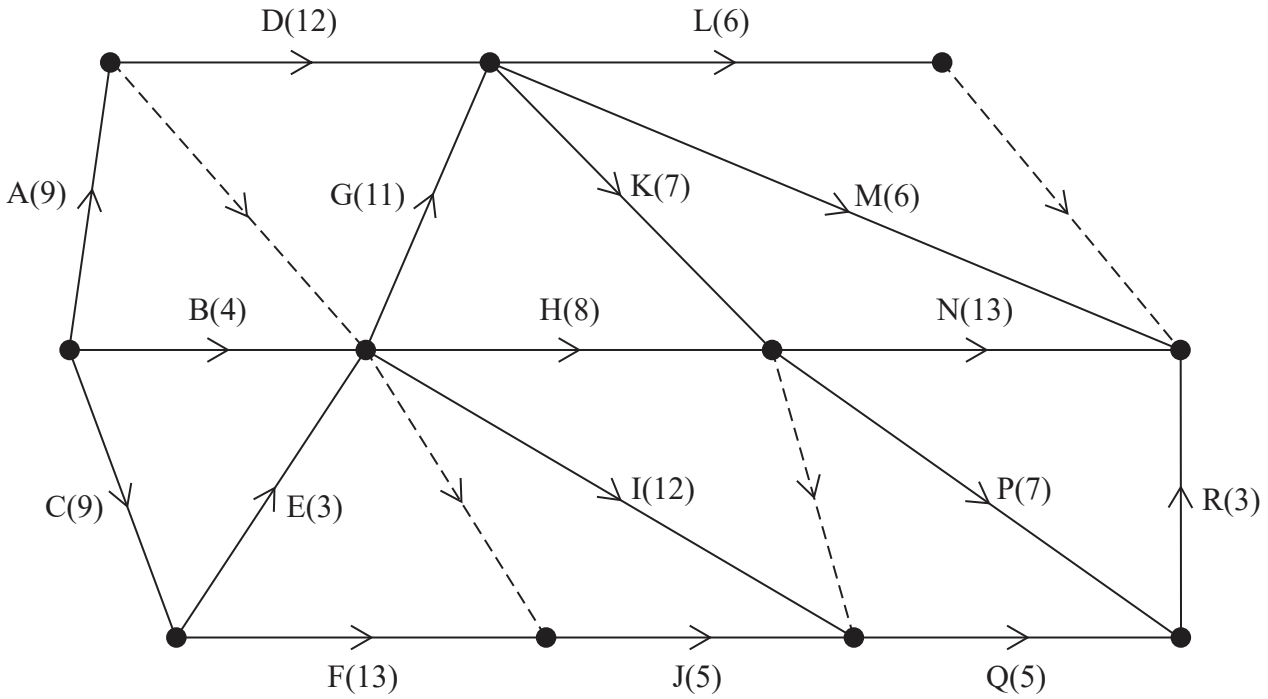


Figure 2

[*The sum of the durations of all the activities is* 133 *days*]

A project is modelled by the activity network shown in Figure 2. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete the activity. Each activity requires one worker. The project is to be completed in the shortest possible time.

(a) Complete the precedence table in the answer book.

(2)

(b) Complete Diagram 1 in the answer book to show the early event times and the late event times.

(4)

(c) State the critical activities.

(1)

(d) Calculate the total float for activity J. You must make the numbers you use in your calculation clear.

(1)

(e) Calculate a lower bound for the number of workers needed to complete the project in the minimum time. You must show your working.

(1)

Diagram 2 in the answer book shows a partly completed scheduling diagram for this project.

(f) Complete the scheduling diagram, using the minimum number of workers, so that the project is completed in the minimum time.

(4)

(Total for Question 4 is 13 marks)

4.

Activity	Immediately preceded by
A	–
B	–
C	–
D	A
E	C
F	C

Activity	Immediately preceded by
G	
H	
I	
J	
K	D, G
L	D, G

Activity	Immediately preceded by
M	D, G
N	
P	
Q	
R	

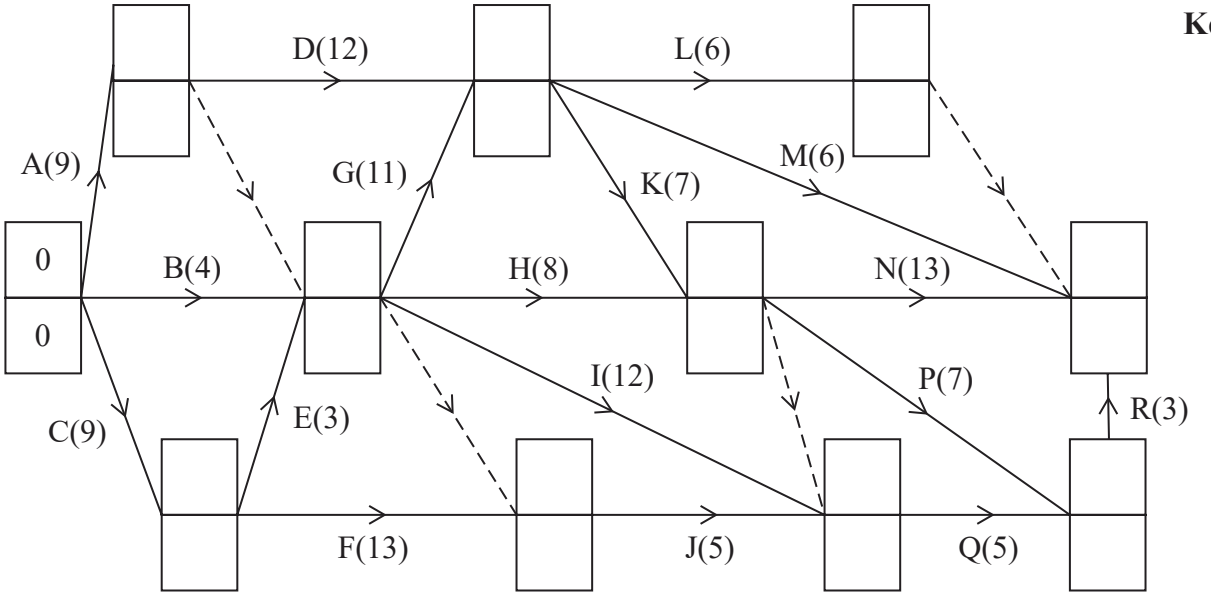


Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 4 continued

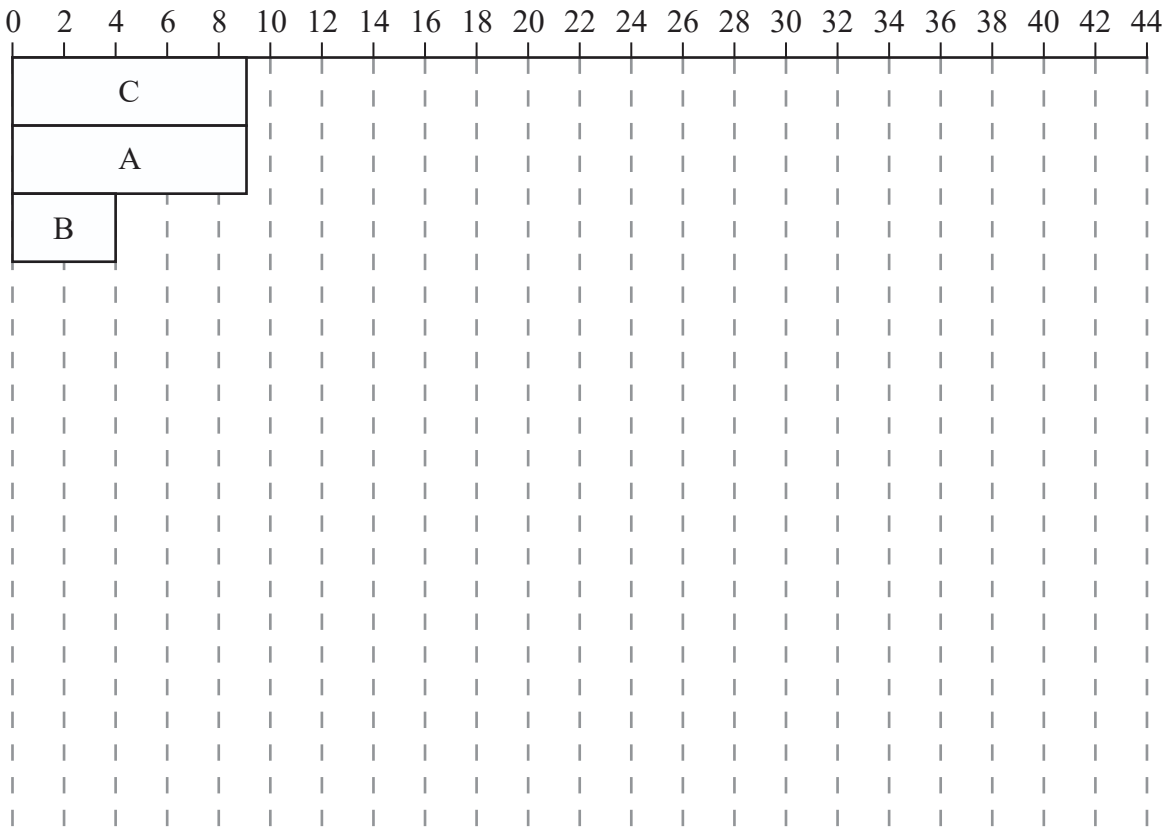


Diagram 2

(Total for Question 4 is 13 marks)

Write your answers in the D1 answer book for this paper.

1.

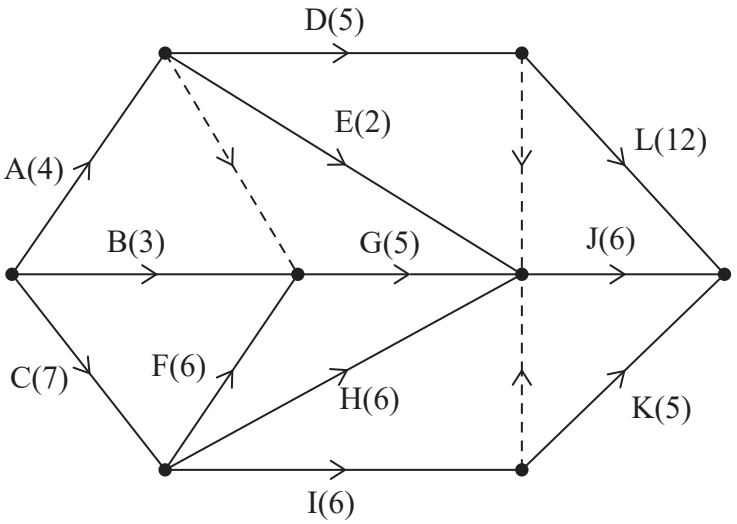


Figure 1

A project is modelled by the activity network shown in Figure 1. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the corresponding activity. Each activity requires one worker. The project is to be completed in the shortest possible time using as few workers as possible.

(a) Complete Diagram 1 in the answer book to show the early event times and the late event times.

(4)

(b) Calculate the total float for activity D. You must make the numbers used in your calculation clear.

(1)

(c) Calculate a lower bound for the minimum number of workers required to complete the project in the shortest possible time. You must show your working.

(2)

(d) Draw a cascade chart for this project on Grid 1 in the answer book.

(4)

(e) Use your cascade chart to determine the minimum number of workers needed to complete the project in the shortest possible time. You must make specific reference to time and activities. (You do not need to provide a schedule of the activities.)

(2)

(Total for Question 1 is 13 marks)

1.

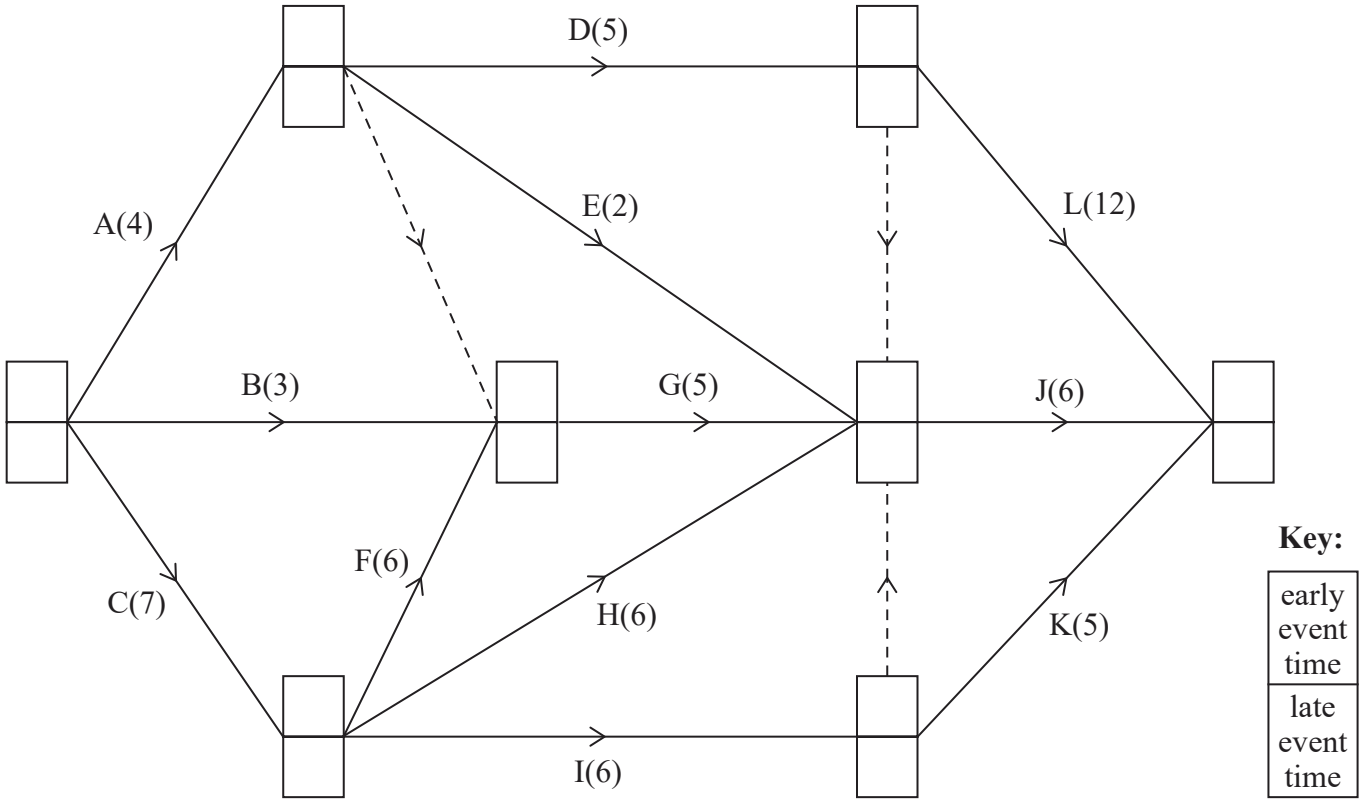
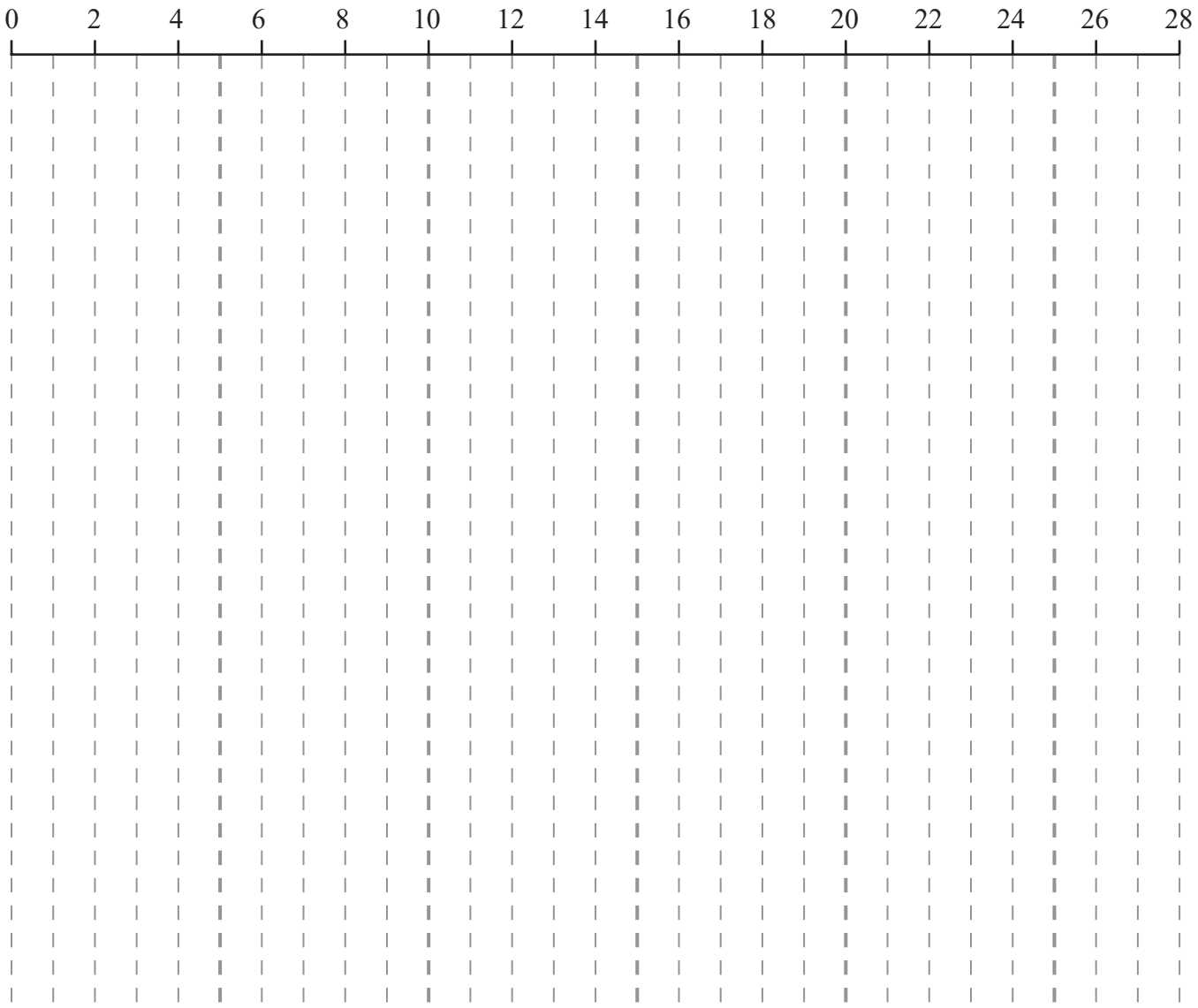


Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 1 continued



Grid 1

(Total for Question 1 is 13 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4.

Activity	Immediately preceding activities
A	–
B	–
C	A, B
D	A, B
E	B
F	C, D, E
G	F
H	B
I	F
J	F
K	G
L	G, H, I, J
M	G, I

(a) Draw the activity network described in the precedence table, using activity on arc and the minimum number of dummies.

(5)

(b) Given that

- the activity network contains only one critical path
- activity E is on this critical path

state

(i) which activities could **never** be critical,

(ii) which activities **must** be critical.

(2)

(Total for Question 4 is 7 marks)

4.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Only use this blank space if you need to redraw your activity network.

(Total for Question 4 is 7 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. (a) Draw the activity network described by the precedence table below, using activity on arc. Use dummies only where necessary.

(5)

Activity	Immediately preceding activities
A	–
B	–
C	A
D	A, B
E	C, D
F	D
G	C
H	G
I	G
J	E, F, I
K	F

Given that K is a critical activity,

(b) state which other activities must also be critical.

(1)

Given instead that all activities shown in the precedence table have the same duration and K is not necessarily critical,

(c) state the critical path for the network.

(1)

(Total 7 marks)

Leave blank

4.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Leave blank

Question 4 continued

Q4

(Total 7 marks)

5.

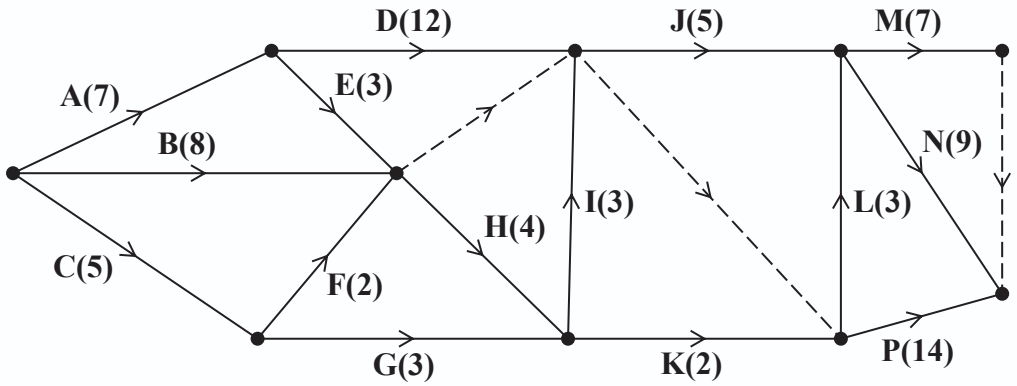


Figure 1

A project is modelled by the activity network shown in Figure 1. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete the corresponding activity. Each activity requires exactly one worker. The project is to be completed in the shortest possible time.

(a) Complete Diagram 1 in the answer book to show the early event times and the late event times.

(4)

(b) Calculate a lower bound for the number of workers needed to complete the project in the minimum time. You must show your working.

(2)

(c) Schedule the activities on Grid 1 in the answer book using the minimum number of workers so that the project is completed in the minimum time.

(4)

Additional resources become available, which can shorten the duration of one of activities D, G or P by one day.

(d) Determine which of these three activities should be shortened to allow the project to be completed in the minimum time. You must give reasons for your answer.

(2)

(Total 12 marks)

Leave blank

5.

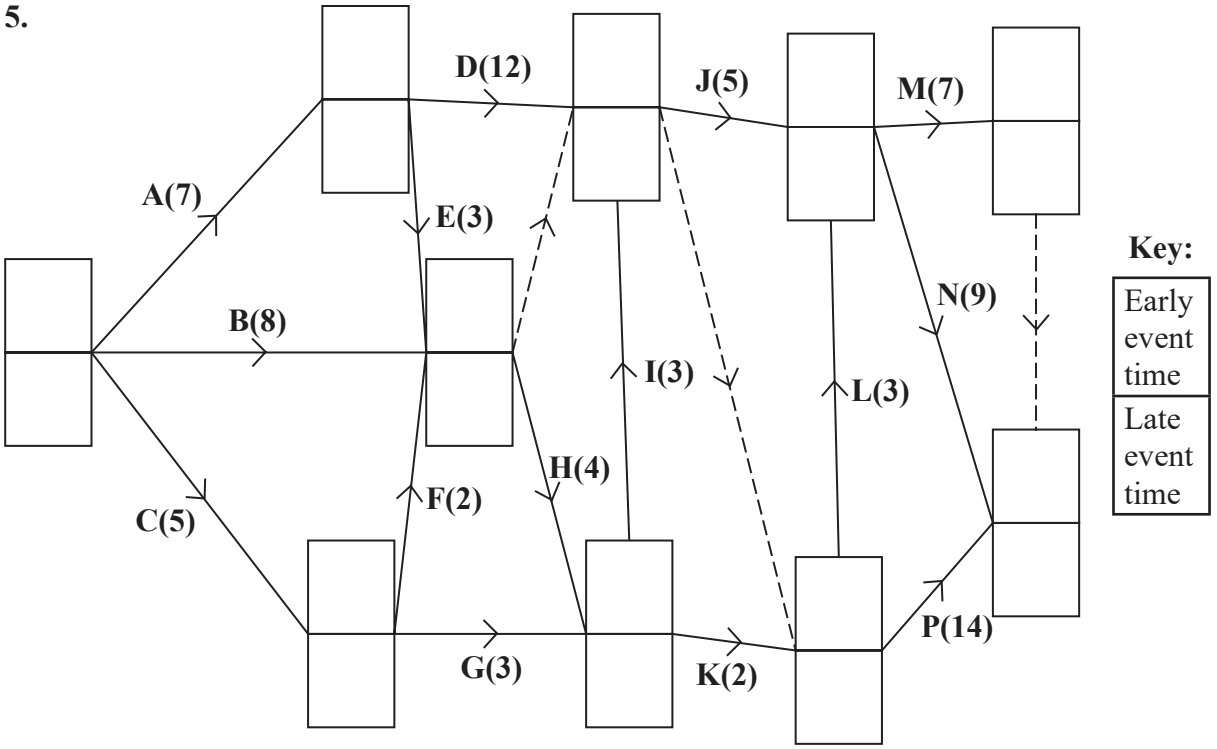


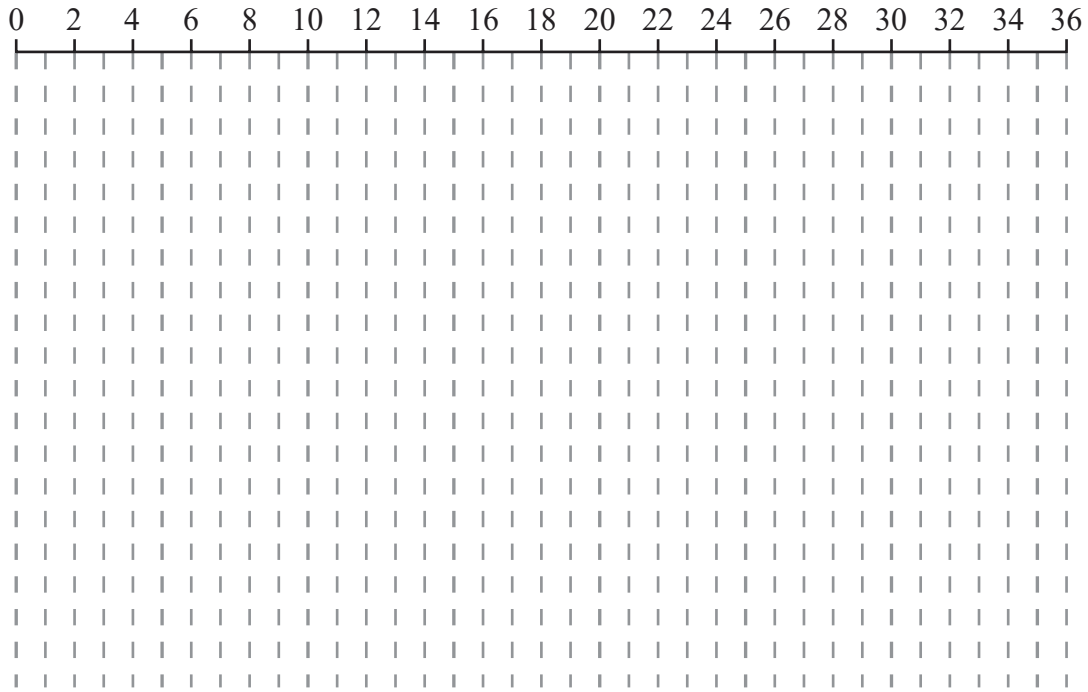
Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Leave blank

Question 5 continued



Grid 1

Q5

(Total 12 marks)

2.

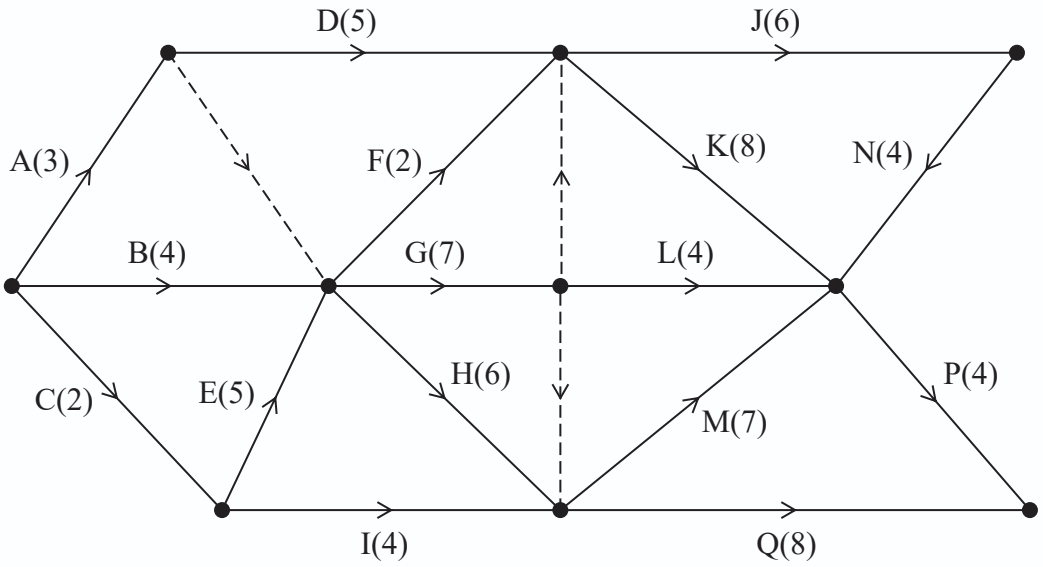


Figure 1

A project is modelled by the activity network shown in Figure 1. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the corresponding activity. Each activity requires one worker. The project is to be completed in the shortest possible time.

(a) Complete Diagram 1 in the answer book to show the early event times and the late event times.

(4)

(b) Draw a cascade chart for this project on Grid 1 in the answer book.

(4)

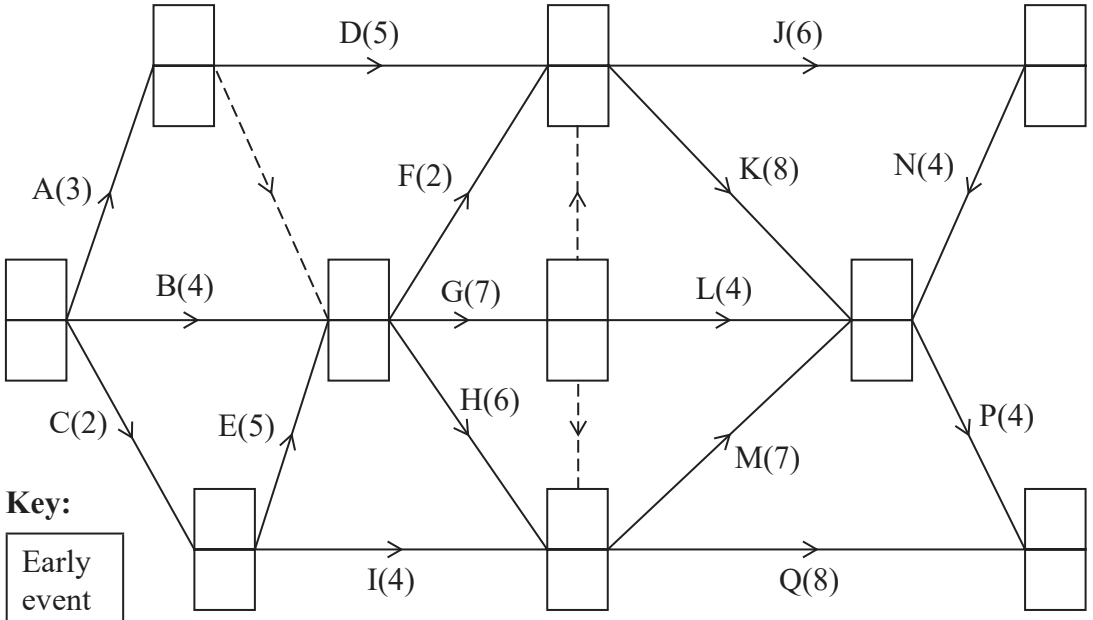
(c) Use your cascade chart to determine the minimum number of workers needed to complete the project in the shortest possible time. You must make specific reference to time and activities. (You do not need to provide a schedule of the activities.)

(2)

(Total 10 marks)

Leave blank

2.



Key:

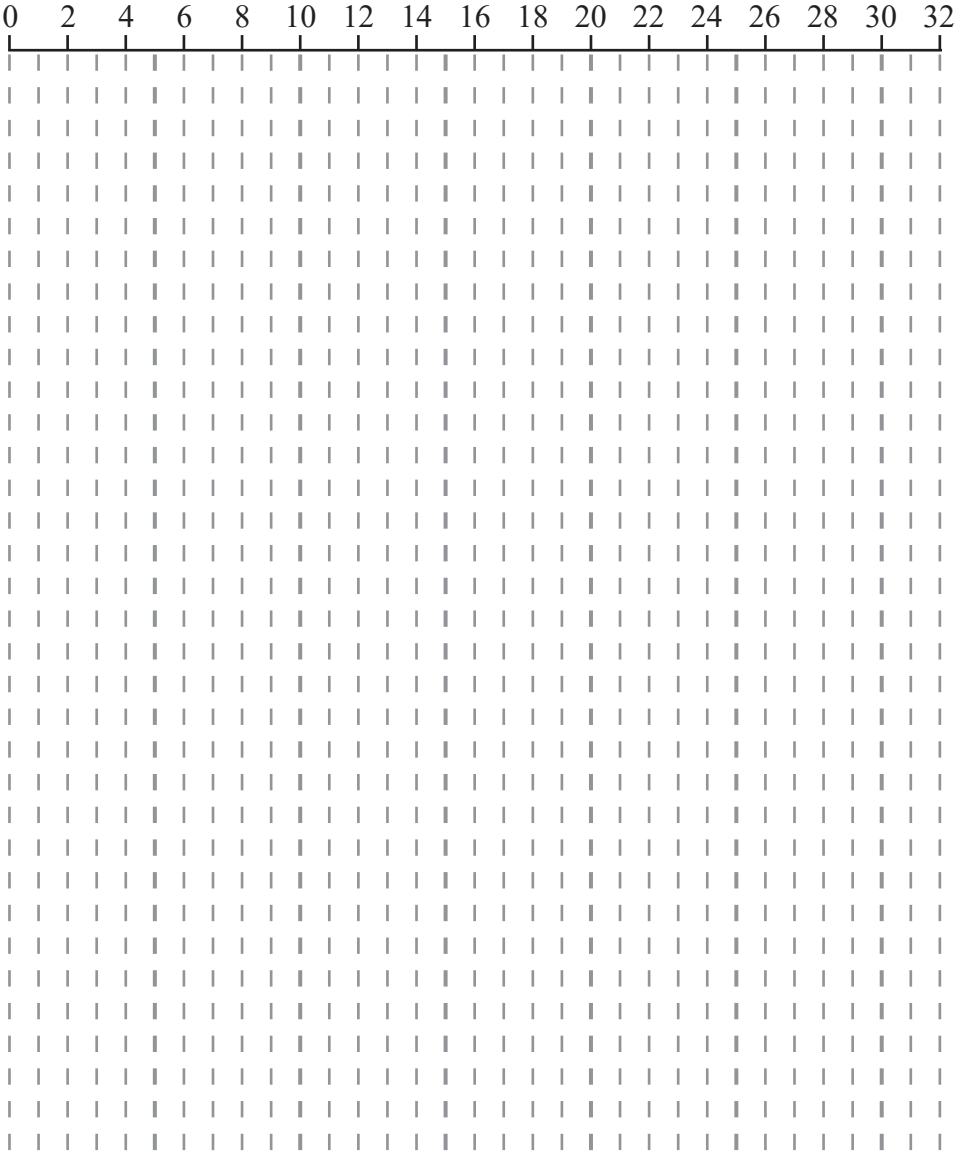
Early event time
Late event time

Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Leave blank

Question 2 continued



Grid 1

Q2

(Total 10 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6.

Activity	Immediately preceding activities
A	–
B	–
C	–
D	A
E	A
F	A, B, C
G	C
H	G
I	D, E, F, H
J	I
K	I
L	I
M	L

(a) Draw the activity network for the project described in the precedence table above, using activity on arc and the minimum number of dummies.

(5)

(b) State which activity is guaranteed to be critical, giving a reason for your answer.

(2)

It is given that each activity in the table takes two hours to complete.

(c) State the minimum completion time and write down the critical path for the project.

(2)

(Total 9 marks)

Leave blank

6.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Leave blank

Question 6 continued

Q6

(Total 9 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2.

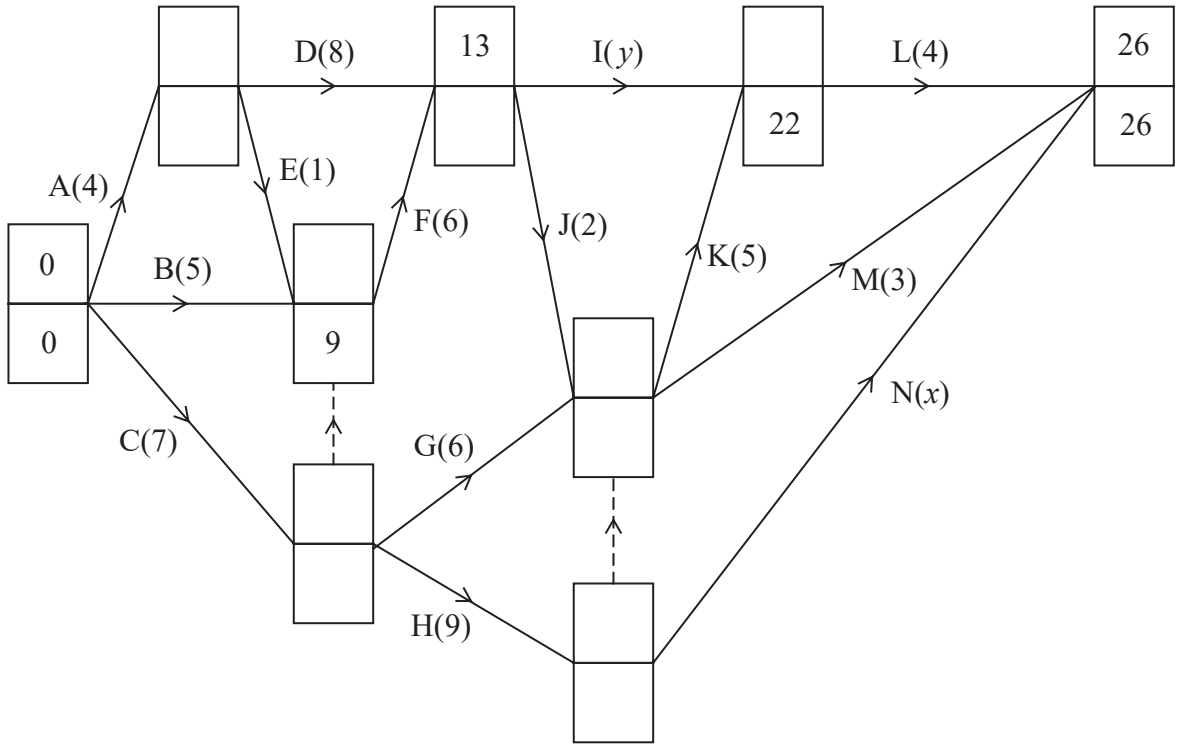


Figure 1

The network in Figure 1 shows the activities that need to be undertaken to complete a project. Each activity is represented by an arc and the duration of the activity, in days, is shown in brackets. The early event times and late event times are to be shown at each vertex and some have been completed.

Given that

- CHN is the critical path for the project
- the total float on activity B is twice the duration of the total float on activity I

(a) find the value of x and show that the value of y is 7

(2)

(b) Calculate the missing early event times and late event times and hence complete Diagram 1 in your answer book.

(3)

Each activity requires one worker, and the project must be completed in the shortest possible time.

(c) Draw a cascade chart for this project on Grid 1 in your answer book, and use it to determine the minimum number of workers needed to complete the project in the shortest possible time. You must make specific reference to time and activities.

(6)

(Total for Question 2 is 11 marks)

2.

(a) _____

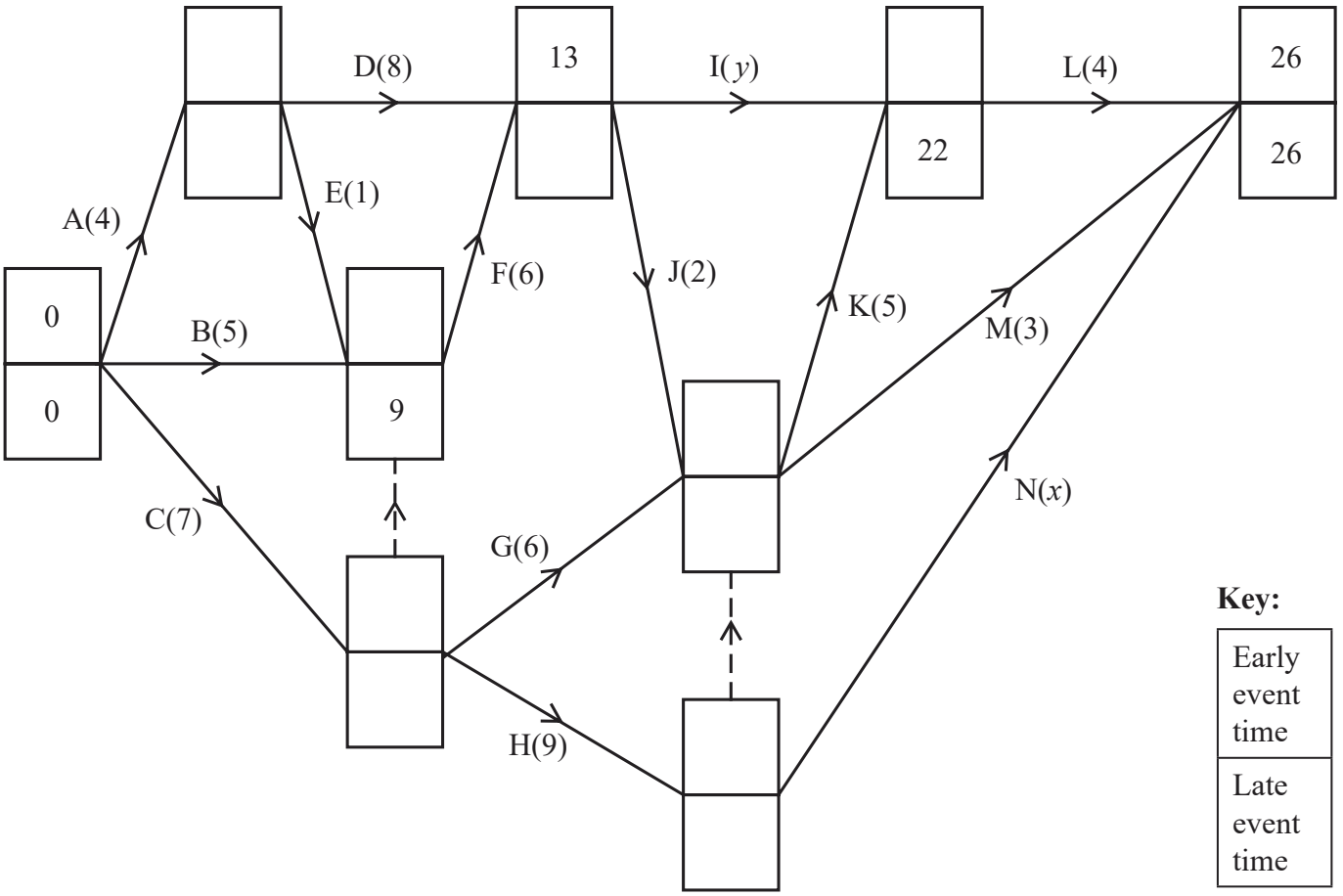
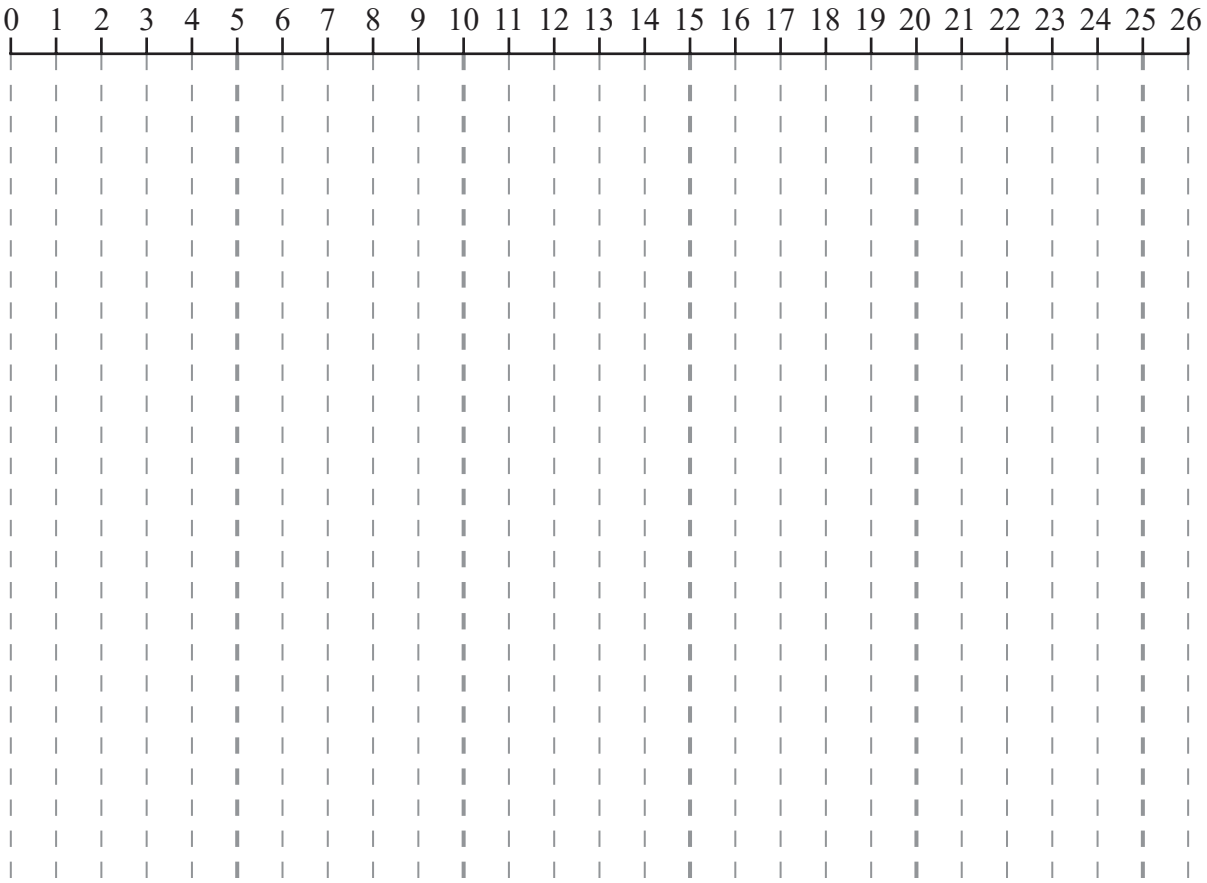


Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 2 continued



Grid 1

(Total for Question 2 is 11 marks)

5. The precedence table shows the eleven activities required to complete a project.

Activity	Immediately preceding activities
A	–
B	–
C	–
D	A, B
E	A, B
F	B, C
G	B, C
H	D
I	D, E, F, G
J	H, I
K	D, E, F

(a) Draw the activity network for the project, using activity on arc and the minimum number of dummies.

(5)

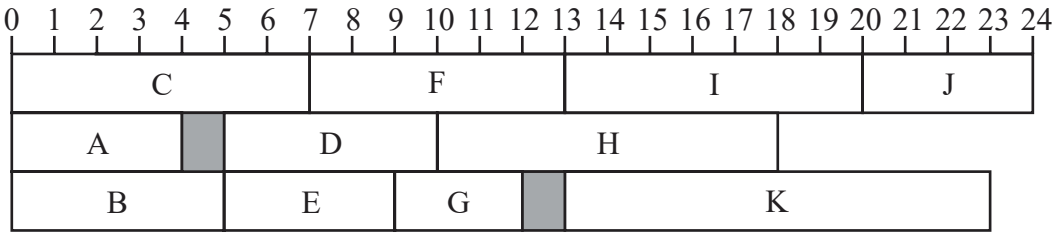


Figure 3

Figure 3 shows a schedule for the project. Each of the activities shown in the precedence table requires one worker. The time taken to complete each activity is in hours and the project is to be completed in the minimum possible time.

(b) (i) State the minimum completion time for the project.

(ii) State the critical activities.

(iii) State the total float on activity G and the total float on activity K.

(4)

(Total for Question 5 is 9 marks)

5.

Activity	Immediately preceding activities
A	–
B	–
C	–
D	A, B
E	A, B
F	B, C
G	B, C
H	D
I	D, E, F, G
J	H, I
K	D, E, F

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 5 continued

Please redraw your activity network on this page if you need to do so.

(Total for Question 5 is 9 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Write your answers in the D1 answer book for this paper.

1.

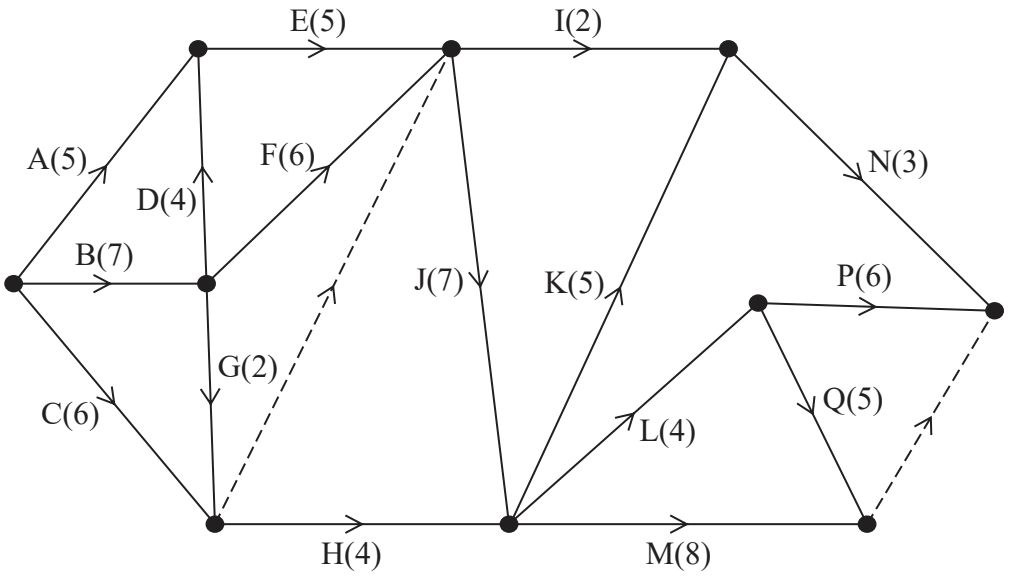


Figure 1

A project is modelled by the activity network shown in Figure 1. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete the activity. Each activity requires exactly one worker. The project is to be completed in the shortest possible time.

(a) Complete Diagram 1 in the answer book to show the early event times and the late event times.

(4)

(b) Calculate the maximum number of days by which activity H could be delayed without lengthening the completion time of the project. You must make the numbers used in your calculation clear.

(1)

(c) Calculate a lower bound for the number of workers needed to complete the project in the minimum time. You must show your working.

(2)

(d) Schedule the activities on Grid 1 in the answer book, using the minimum number of workers, so that the project is completed in the minimum time.

(3)

(Total for Question 1 is 10 marks)

1.

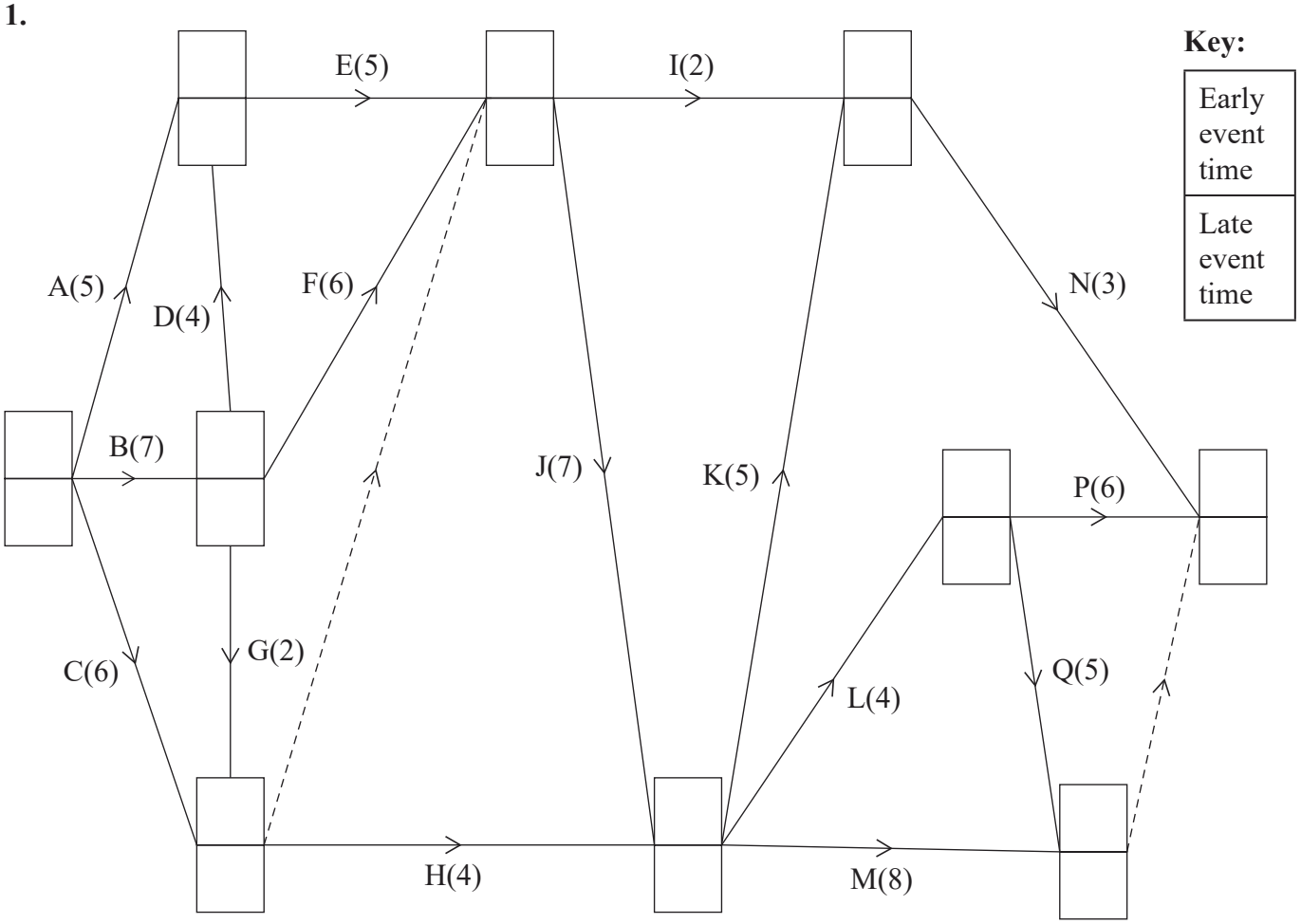
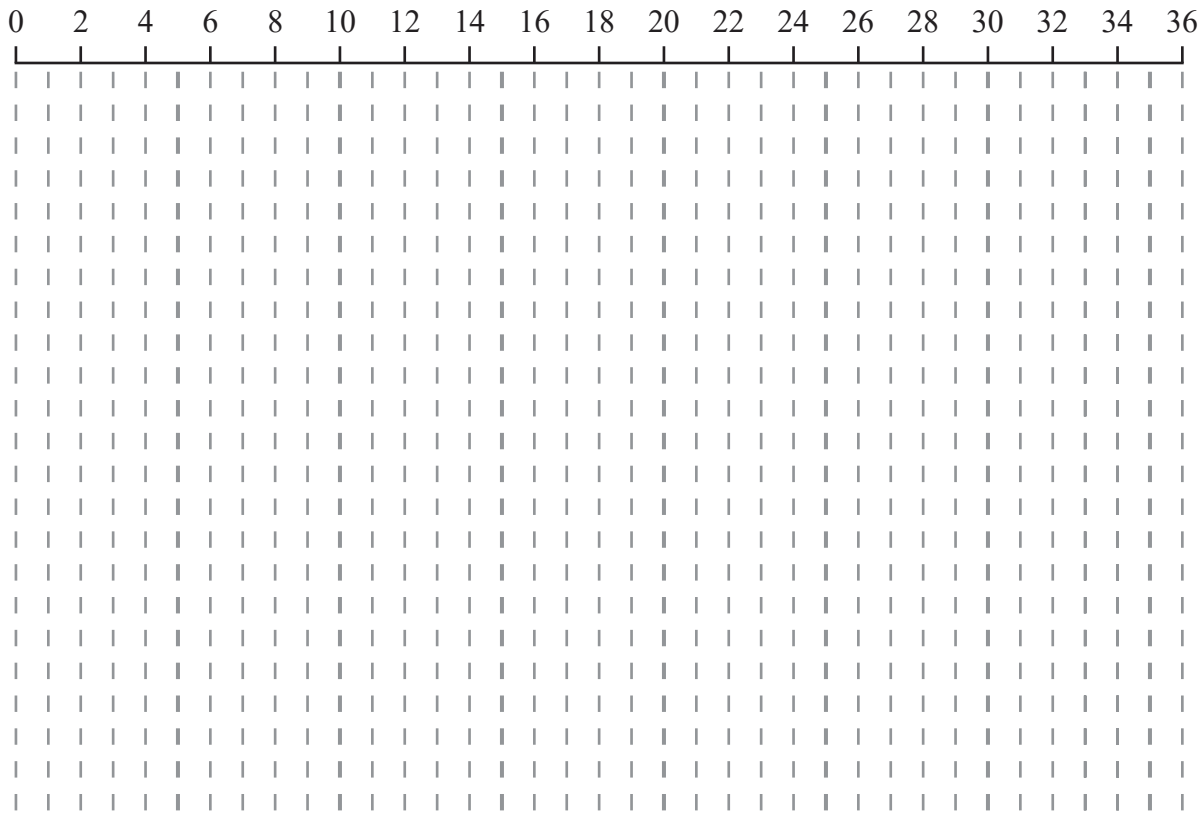


Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 1 continued



Grid 1

(Total for Question 1 is 10 marks)

5.

Activity	Immediately preceding activities
A	-
B	-
C	-
D	A
E	A
F	B, C, E
G	B, C, E
H	C
I	C
J	D, F, G, H, I
K	D, F, G, H, I
L	I

(a) Draw the activity network described in the precedence table above, using activity on arc and the minimum number of dummies.

(5)

A project is modelled by the activity network drawn in (a). Each activity requires exactly one worker. The project is to be completed in the shortest possible time. The table below gives the time, in hours, to complete three of the activities.

Activity	Duration (in hours)
A	10
E	7
F	8

The length of the critical path AEFK is 33 hours.

(b) Determine the range of possible values for the duration of activity J. You must make your method and working clear.

(2)

(Total for Question 5 is 7 marks)

5.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 5 continued

(Total for Question 5 is 7 marks)

2.

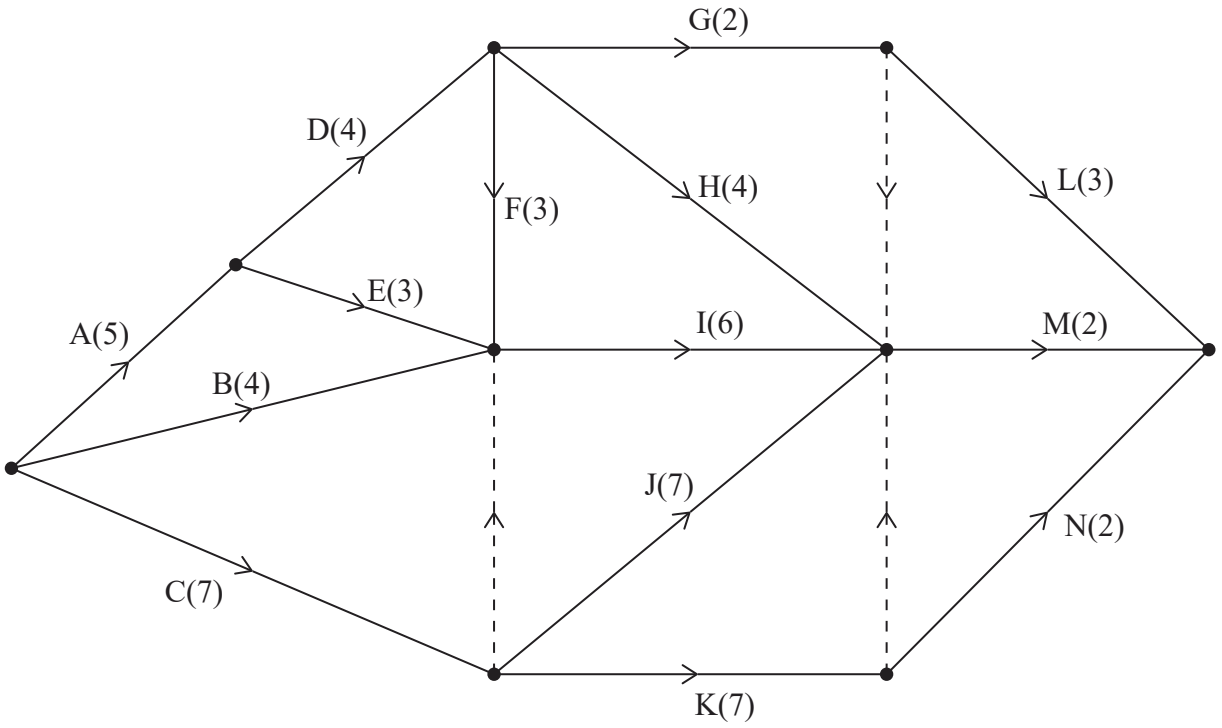


Figure 1

[*The sum of the durations of all the activities is* 59 *days*.]

The network in Figure 1 shows the activities that need to be undertaken to complete a project. Each activity is represented by an arc and the duration, in days, of the corresponding activity is shown in brackets. Each activity requires one worker. The project is to be completed in the shortest possible time.

(a) (i) Complete Diagram 1 in the answer book to show the early event times and the late event times.

(ii) State the minimum completion time of the project.

(5)

(b) Calculate a lower bound for the number of workers needed to complete the project in the minimum time. You must show your working.

(1)

(c) Schedule the activities using the minimum number of workers so that the project is completed in the minimum time.

(4)

(Total for Question 2 is 10 marks)

2.

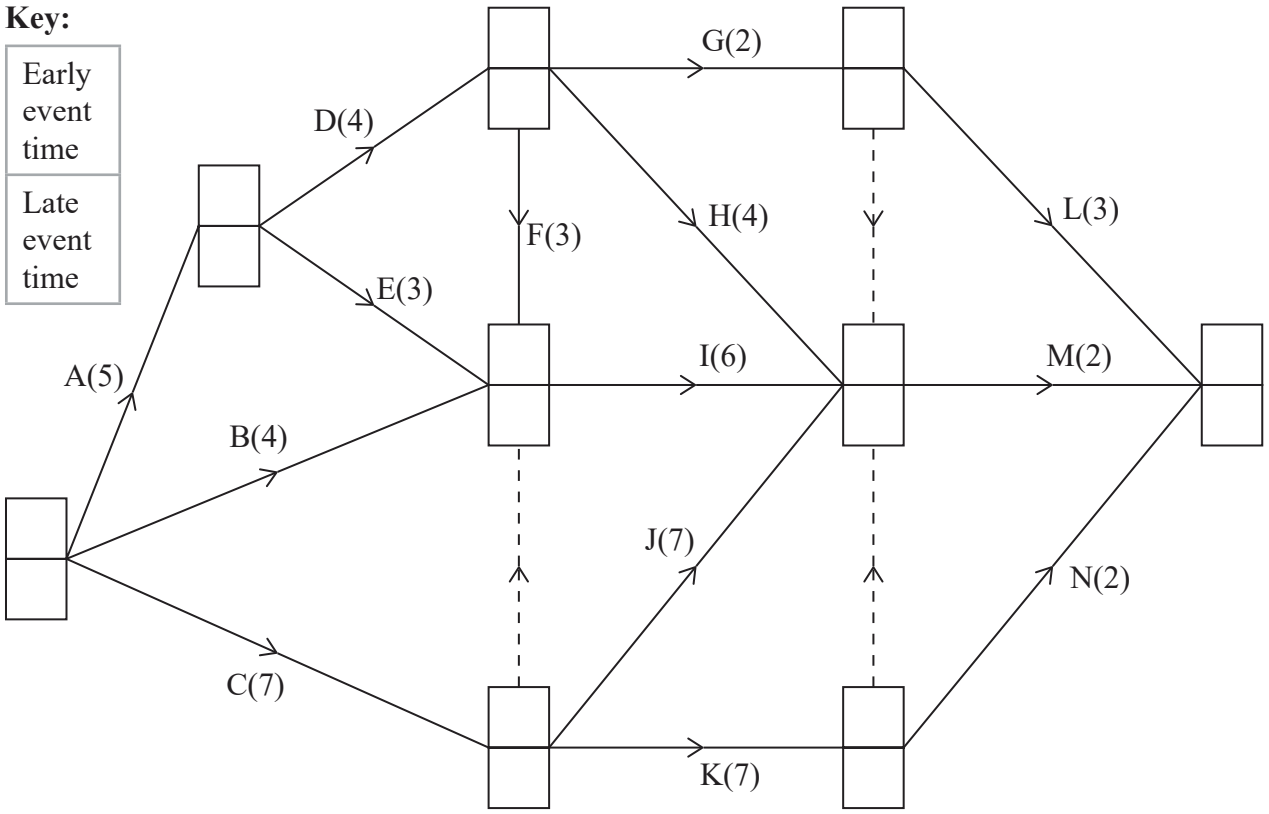
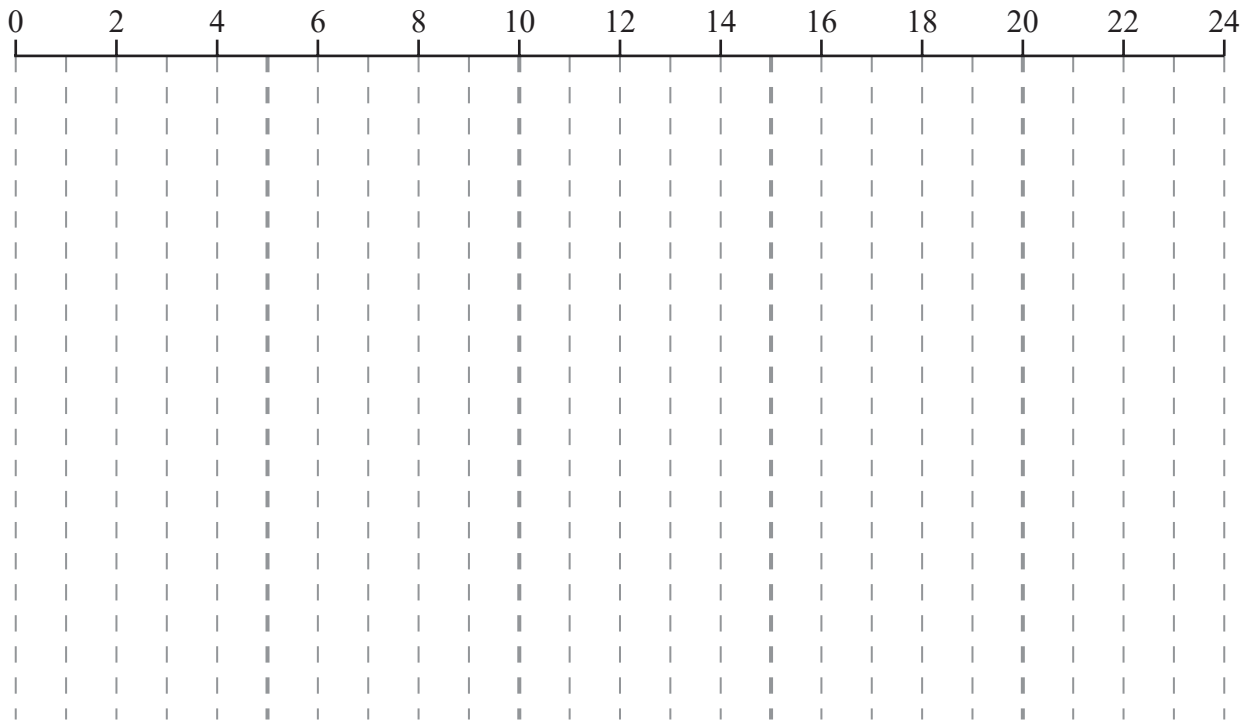


Diagram 1

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 2 continued



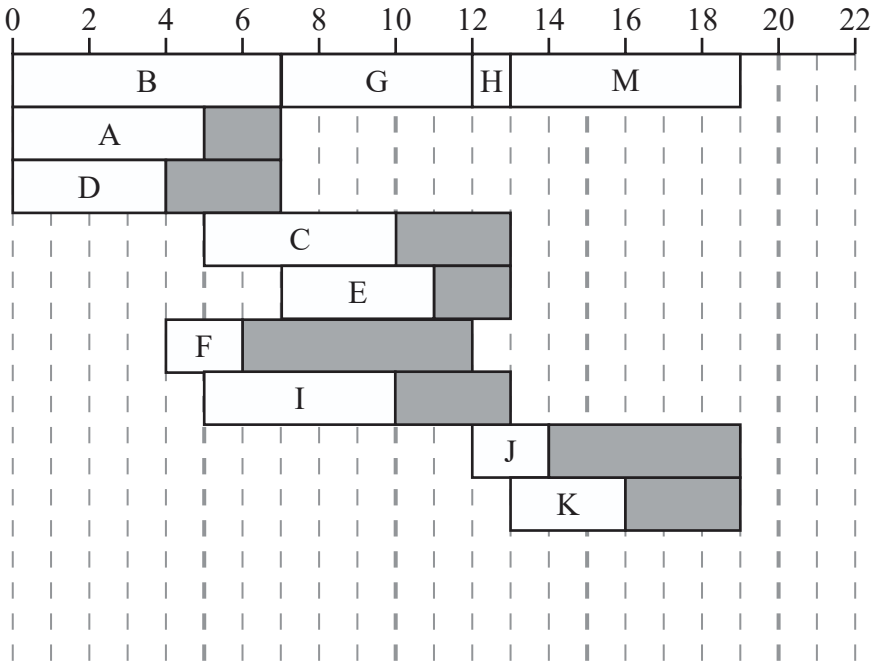
(Total for Question 2 is 10 marks)

6.

Activity	Immediately preceding activities
A	-
B	-
C	A
D	-
E	A, B, D
F	D
G	A, B, D
H	F, G
I	A
J	F, G
K	C, E, H, I
L	I
M	C, E, H, I

(a) Draw the activity network for the project described in the precedence table, using activity on arc and the minimum number of dummies.

(5)



Grid 1

A cascade chart for all the activities of the project, **except activity L**, is shown on Grid 1.

The time taken to complete each activity is given in hours and each activity requires one worker.

The project is to be completed in the minimum time using as few workers as possible.

(b) State the critical activities of the project.

(1)

(c) Use the cascade chart to determine the minimum number of workers needed to complete the project in the shortest possible time. You must make specific reference to time and activities. (You do **not** need to provide a schedule of the activities.)

(2)

The duration of activity L is x hours. Given that the total float of activity L is at most 7 hours,

(d) determine the range of possible values for x.

(2)

(Total for Question 6 is 10 marks)

6.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 6 continued

(Total for Question 6 is 10 marks)

4.

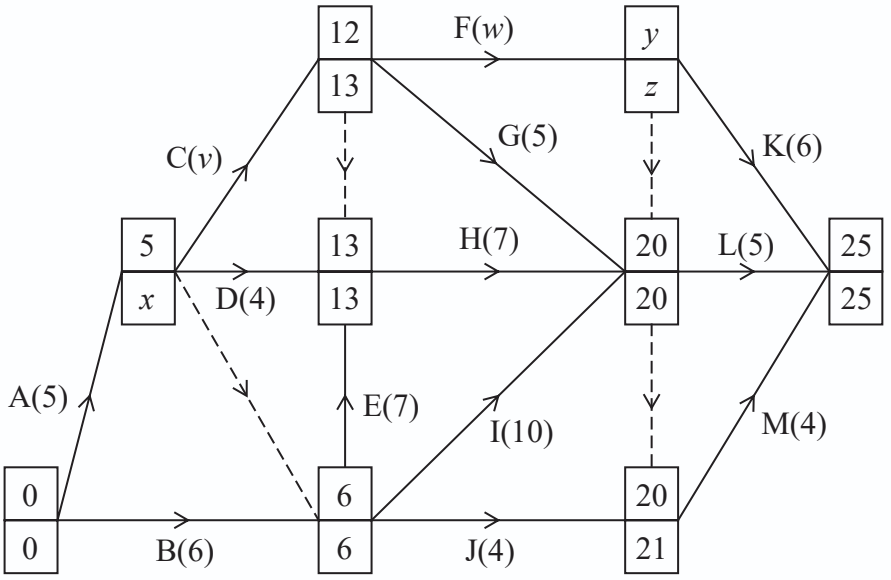


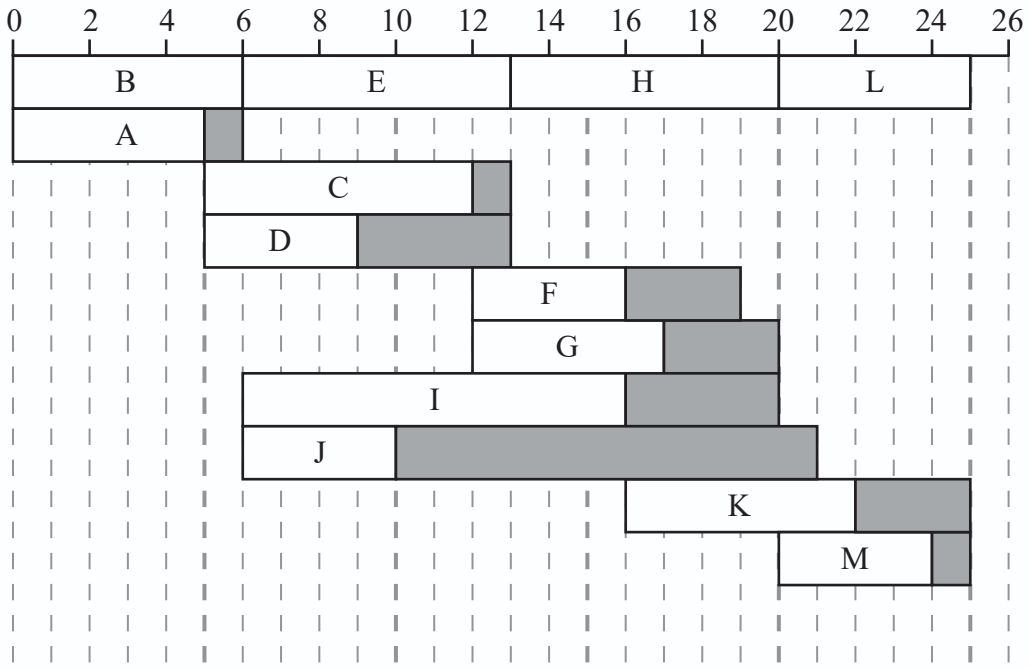
Figure 2

The network in Figure 2 shows the activities that need to be carried out by a company to complete a project. Each activity is represented by an arc, and the duration, in days, is shown in brackets. Each activity requires one worker. The early event times and the late event times are shown at each vertex.

(a) Complete the precedence table in the answer book.

(2)

A cascade chart for this project is shown on Grid 1.



Grid 1

(b) Use Figure 2 and Grid 1 to find the values of v, w, x, y and z.

(3)

The project is to be completed in the minimum time using as few workers as possible.

(c) Calculate a lower bound for the minimum number of workers required. You must show your working.

(1)

(d) On Grid 2 in your answer book, construct a scheduling diagram for this project.

(3)

Before the project begins it is found that activity F will require an additional 5 hours to complete. The durations of all other activities are unchanged. The project is still to be completed in the shortest possible time using as few workers as possible.

(e) State the new minimum project completion time and state the new critical path.

(2)

(Total 11 marks)

Leave blank

4.

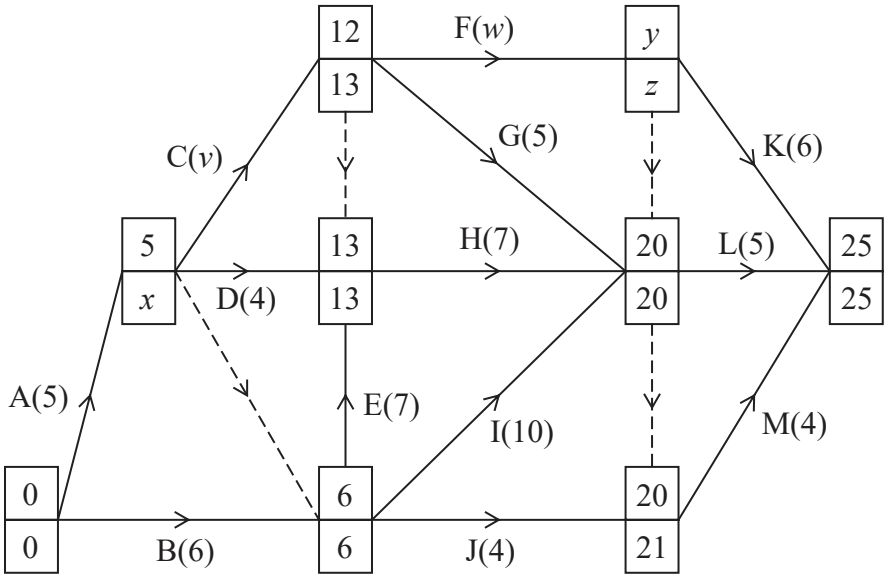


Figure 2

(a)

Activity	Immediately preceding activities
A	
B	
C	
D	
E	

Activity	Immediately preceding activities
F	
G	
H	
I	
J	

Activity	Immediately preceding activities
K	
L	
M	

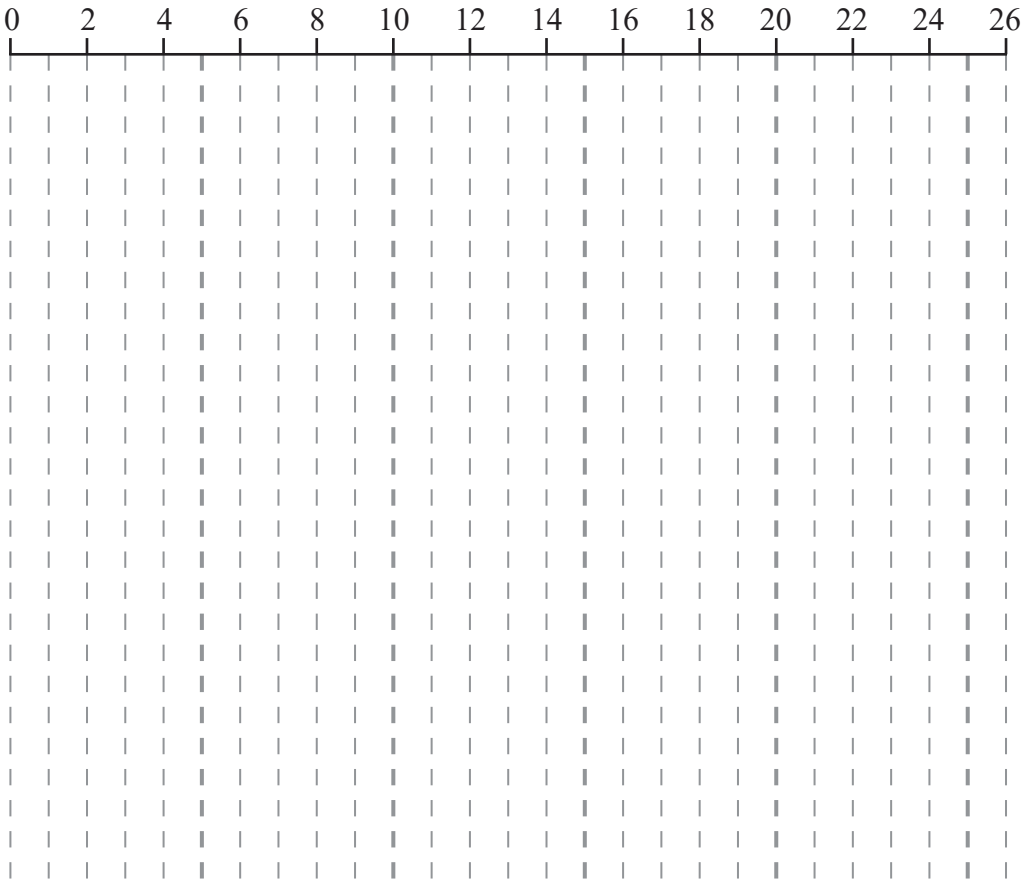
$v =$ _____ $w =$ _____ $x =$ _____ $y =$ _____ $z =$ _____

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Leave blank

Question 4 continued



Grid 2

Q4

(Total 11 marks)